AI Tools in Economics Research

Working with Claude Code and Codex

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Context and the next token

Chat A

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You: `The Queen is`

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Assistant: ... Elizabeth.

You: `Describe the role`

Assistant: ... the head of state.

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The harness (Body)

- Provides access to files and tools.
- Manages context and permissions.
- Returns tool results to the model.

Instructions to the Brain (CLAUDE.md)

```
# Role
You are an Economics PhD student at the University of Chicago...

# Attitude
You should always be skeptical and constructive in your suggestions...

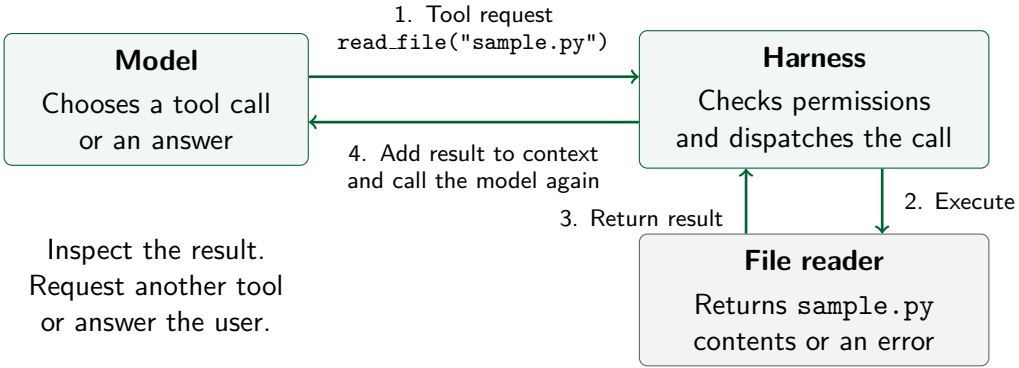
# Writing Style
You must write according to ‘writing_style.md‘.
For research advising and feedback, use ‘JesseShapiroStyle.md‘...

# Basic Rules
- Assume large data size and always prefer duckdb, polars...

# Critical Rules
- NEVER use ‘rm‘ or ‘rm -rf‘ under any circumstances.
- If asked to delete a file, you must instead simply rename the file...
```

The model–harness tool loop

User: Why did the sample shrink? Check `sample.py`.



A tool result becomes new context for the next model output.

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Because of next token prediction, LLMs tend to go down a rabbit hole.

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Inputs	A research use	Model examples
Image + text	Discuss a regression-table screenshot.	GPT, Claude, Gemini; **Qwen3.8, DeepSeek V4.1, Gemma 4**
Audio + text	Analyze a short interview recording.	Gemini, GPT audio models; **Gemma 4 (E2B/E4B/12B)**
Video + text	Review a short presentation clip.	Gemini; **Qwen3.8, Gemma 4**

Bold: open-weight models.

Check both model support and the inputs your harness accepts.

Examples of agent harnesses

The harness supplies the interface, tools, and workflow around the model.

Harness	Where you work	Workflow emphasis
Claude Code	Terminal, editor, app	Project files and command tools.
Codex	App, terminal, editor	Local work and cloud tasks.
Zoo Code	VS Code extension	Model-provider choice and task modes.
Aider	Terminal, app	Code edits with Git-tracked changes.
Hermes Agent	Gateways (e.g. discord)	Private assistant on computer.

Skills and making your own

A **skill** packages a repeatable task as instructions the harness can load when needed.

Example: application review

When: tailor or audit an application.

Inputs: posting, CV, verified facts.

Output: checklist, issues, tailored draft.

Checks: trace claims to the records.

Making your own

1. Capture a workflow that worked.
2. Write `SKILL.md`: when to use it, steps, outputs, and checks.
3. Test on a new case and revise.

Templates, reference files, and helper scripts are optional.

A first workflow to try

1. Choose one task you already know how to evaluate.
2. Provide a small folder and identify the authoritative inputs.
3. Specify a reviewable output and an observable check.
4. Inspect the result, then save the useful instructions for next time.

YOU define the question and judges the evidence.

Questions?